Requirements Verification

Evidence for each of the 11 project requirements, with status and screenshots.

Blake Otto · Catalyst Switch Lab — Small Office Network Build
github.com/blakeotto57/Catalyst-Switch-Lab-Small-Office-Network-Build

Requirement verification

Evidence for each requirement, extracted from my documented project brief. Screenshots are in `images/`.

#	Requirement	Status
1	Clean baseline	Met
2	Secure management	Met (SW3: no SSH support, so all remote access is disabled)
3	Segmentation	Met
4	Redundant core uplinks	Met with limitation — SW2 uses LACP; SW3 (3548 XL) has no LACP, so its uplink is a static EtherChannel
5	Access cross-link	Met
6	Spanning tree	Met with limitation — Rapid PVST+ on R1 and SW2; SW3 (3548 XL) supports only 802.1D
7	Centralized routing	Met
8	DHCP	Met
9	Logging and backups	Met
10	Internet access	Partially met — documented limitation
11	Access-layer security (optional)	Met on SW2; SW3 (3548 XL) does not support DHCP snooping and has no VLAN 10 user ports

Requirement 1: Clean baseline

Accepted when: All three switches are inventoried (model, software, capabilities) and reset to a blank configuration

Status: Met

> Inventory evidence (`show version` for each switch) is not in the document yet — add it.

Evidence and notes

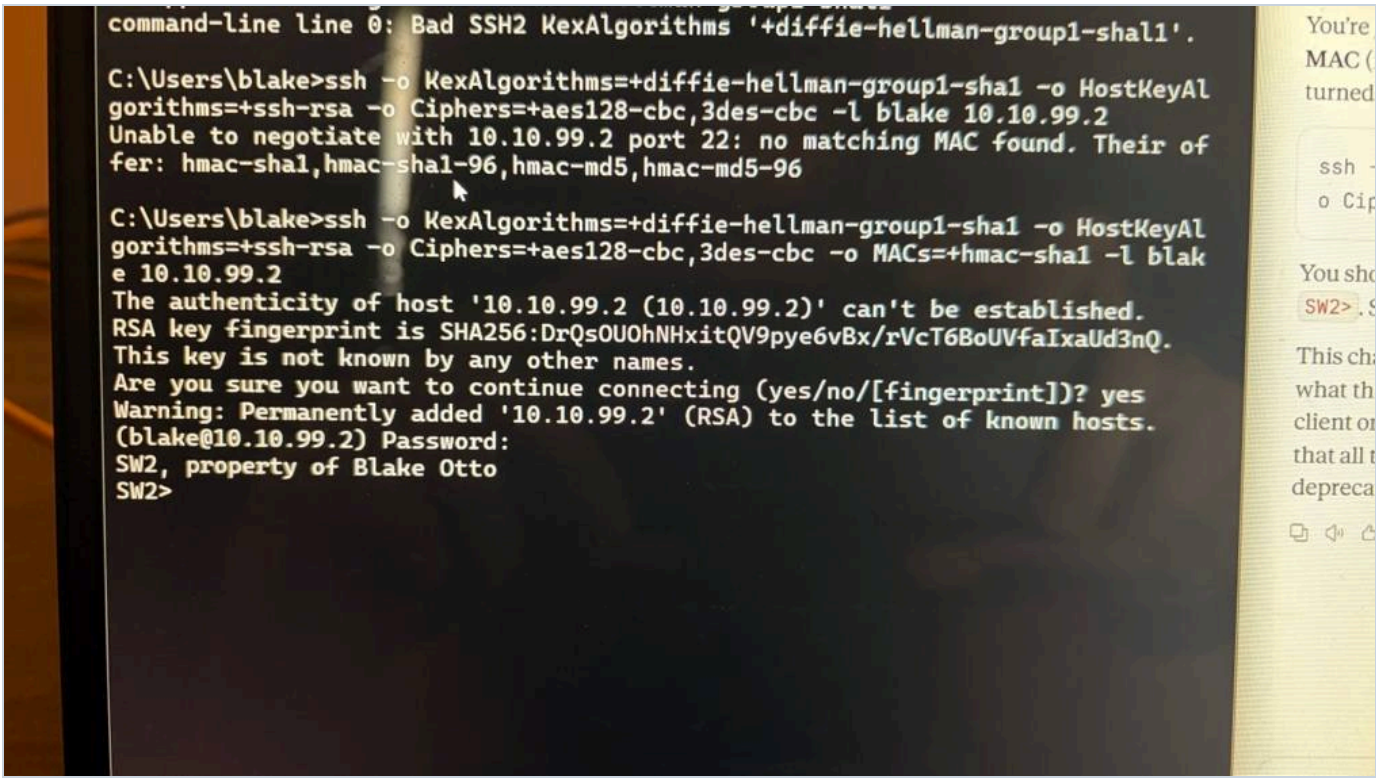
Requirement 2: Secure management

Accepted when: Every switch has a hostname, encrypted privileged password, local user login and banner; remote access is SSH version 2 only and Telnet is refused

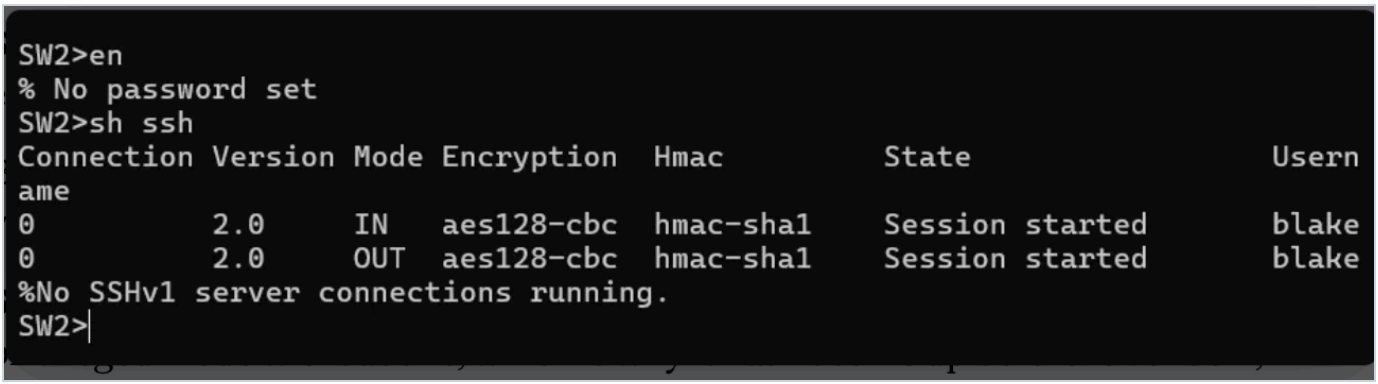
Status: Met (SW3: no SSH support, so all remote access is disabled)

Evidence and notes

We can see SSH works



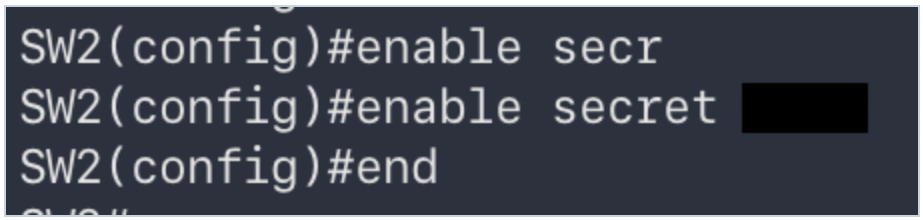
and SSH is ver 2:



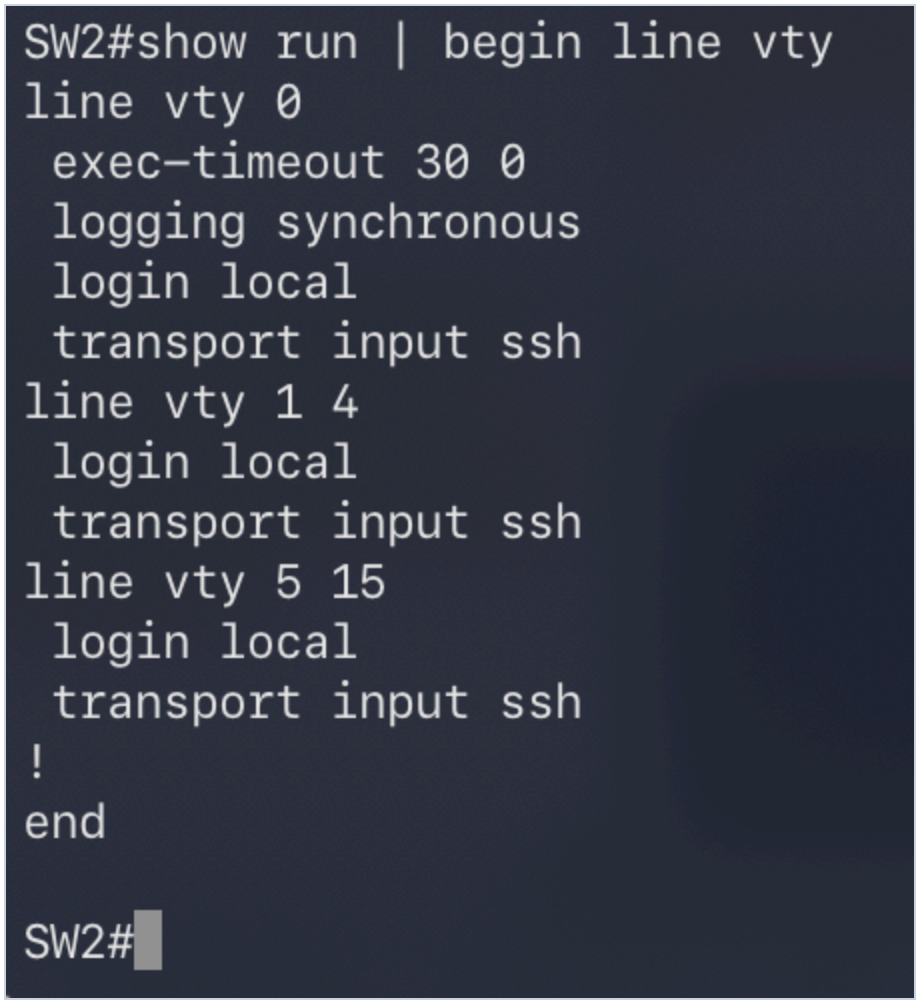
Fixed SW2's missing password on SW2 via:

enable secret
end

Final running config file will be incorrect:



Also Telnet can go through since I did “line vty 0 4” and not “line vty 0 15”. Correction on SW2 and R1:



SW3 cannot do SSH so I restricted remote access:

```
SW3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SW3(config)#line vty 0 15
SW3(config-line)#transport input ssh
                                 ^
% Invalid input detected at '^' marker.

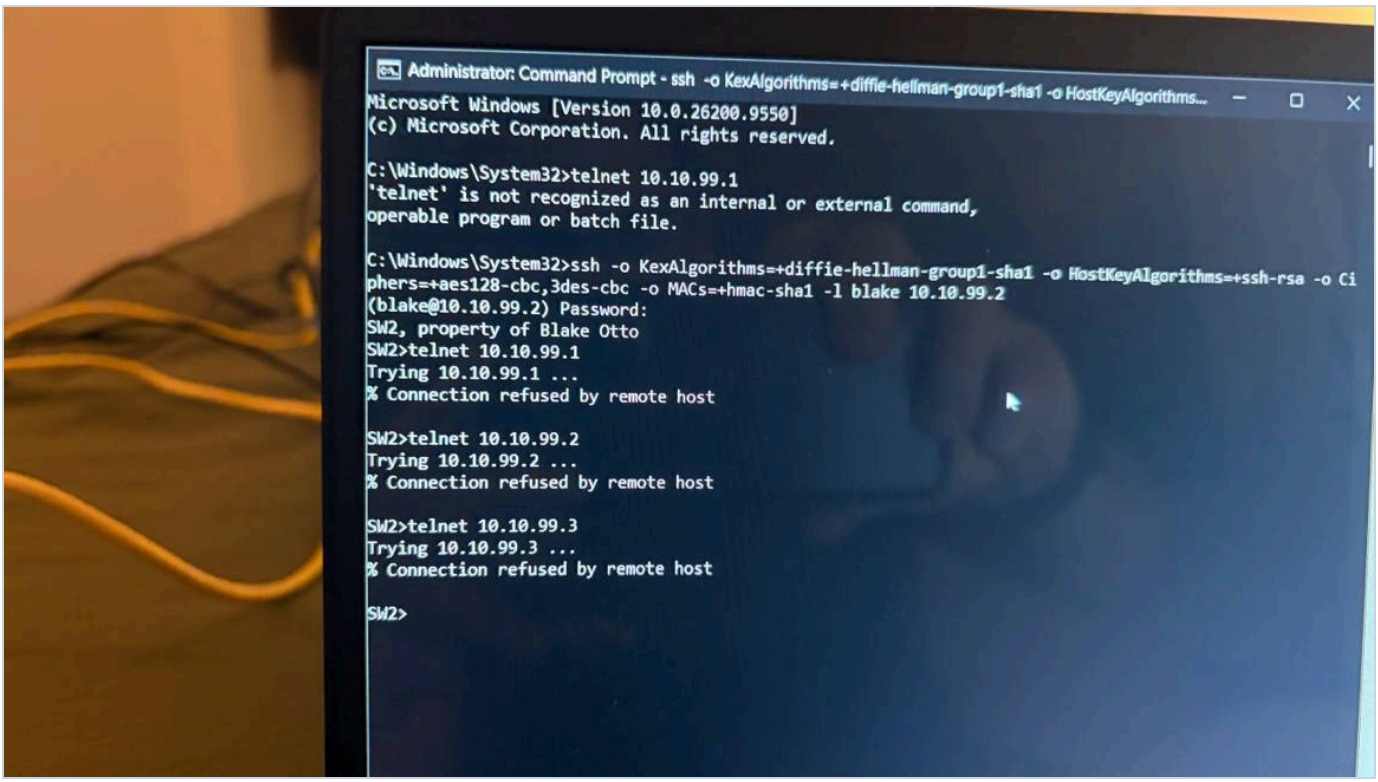
SW3(config-line)#transport input ?
  all     All protocols
  none    No protocols
  telnet  TCP/IP Telnet protocol

SW3(config-line)#transport input none
SW3(config-line)#login local
SW3(config-line)#end
SW3#c
```

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#line vty 0 15
R1(config-line)#trans
R1(config-line)#transport in
R1(config-line)#transport input ssh
R1(config-line)#login local
R1(config-line)#end
R1#s ru
*Mar  1 14:35:48.779: %SYS-5-CONFIG_I: Configured from console by blake on console
R1#sh run | begin line vty
line vty 0
 exec-timeout 30 0
 logging synchronous
 login local
 transport input ssh
line vty 1 4
 login local
 transport input ssh
line vty 5 15
 login local
 transport input ssh
!
end

R1#
```

Telnet is refused on all 3 devices



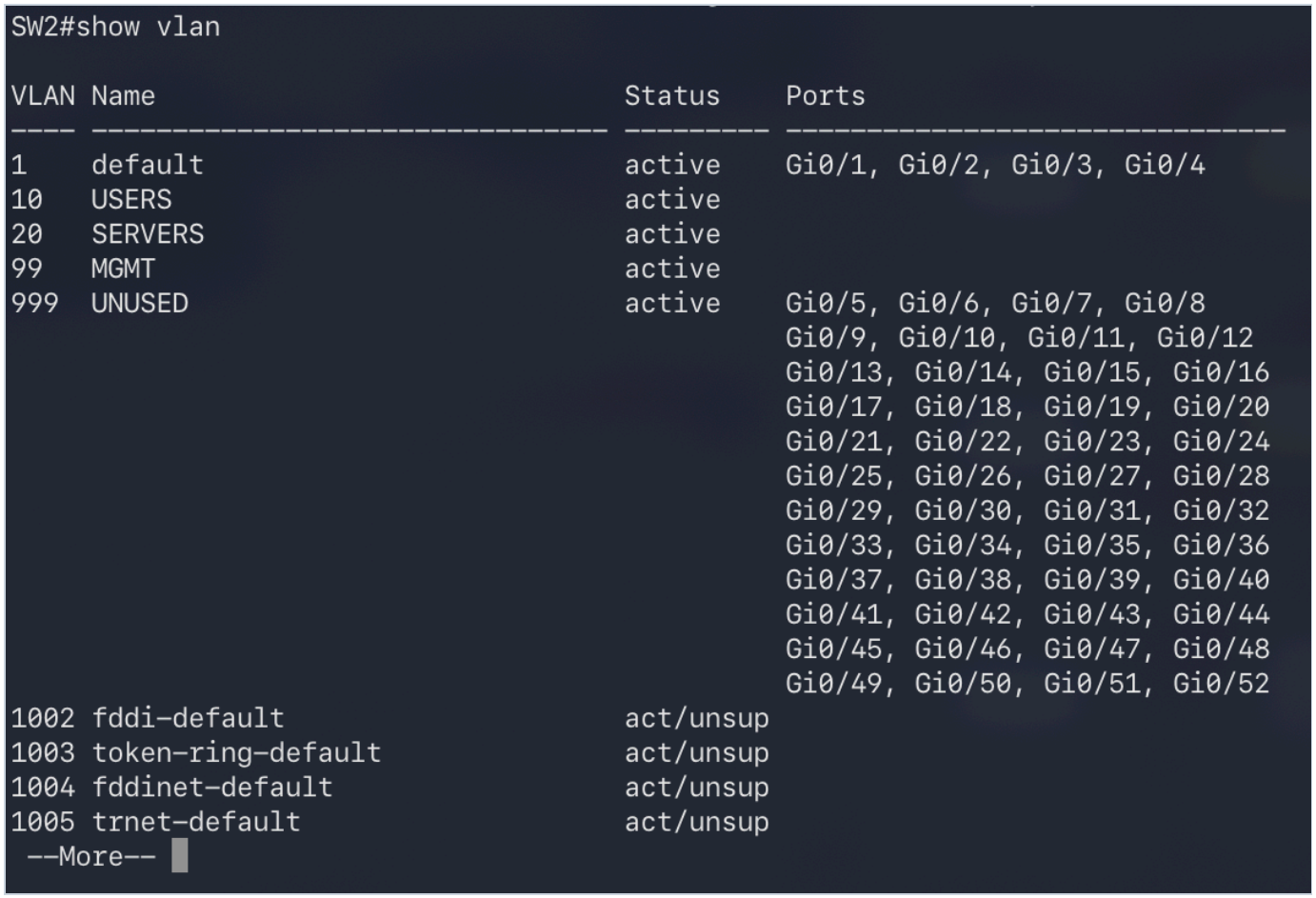
All devices have encrypted secrets:
Can’t show for privacy sake

Requirement 3: Segmentation

Accepted when: VLANs 10, 20 and 99 exist on all three switches; every unused port is in VLAN 999 and shut down

Status: Met

Evidence and notes



```
SW2#show vlan

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Gi0/1, Gi0/2, Gi0/3, Gi0/4
10   USERS                            active
20   SERVERS                          active
99   MGMT                             active
999  UNUSED                           active    Gi0/5, Gi0/6, Gi0/7, Gi0/8
                                                Gi0/9, Gi0/10, Gi0/11, Gi0/12
                                                Gi0/13, Gi0/14, Gi0/15, Gi0/16
                                                Gi0/17, Gi0/18, Gi0/19, Gi0/20
                                                Gi0/21, Gi0/22, Gi0/23, Gi0/24
                                                Gi0/25, Gi0/26, Gi0/27, Gi0/28
                                                Gi0/29, Gi0/30, Gi0/31, Gi0/32
                                                Gi0/33, Gi0/34, Gi0/35, Gi0/36
                                                Gi0/37, Gi0/38, Gi0/39, Gi0/40
                                                Gi0/41, Gi0/42, Gi0/43, Gi0/44
                                                Gi0/45, Gi0/46, Gi0/47, Gi0/48
                                                Gi0/49, Gi0/50, Gi0/51, Gi0/52
1002 fddi-default                     act/unsup
1003 token-ring-default               act/unsup
1004 fddinet-default                  act/unsup
1005 trnet-default                    act/unsup
 --More--
```

```
SW3#show vlan
VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Gi0/1, Gi0/2
10   USERS                            active
20   SERVERS                          active
99   MGMT                             active
999  UNUSED                           active    Fa0/3, Fa0/4, Fa0/5, Fa0/6,
                                                Fa0/7, Fa0/8, Fa0/9, Fa0/10,
                                                Fa0/11, Fa0/12, Fa0/13, Fa0/14,
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18,
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22,
                                                Fa0/23, Fa0/24, Fa0/25, Fa0/26,
                                                Fa0/27, Fa0/28, Fa0/29, Fa0/30,
                                                Fa0/31, Fa0/32, Fa0/33, Fa0/34,
                                                Fa0/35, Fa0/36, Fa0/37, Fa0/38,
                                                Fa0/39, Fa0/40, Fa0/41, Fa0/42,
                                                Fa0/43, Fa0/44, Fa0/45, Fa0/46,
                                                Fa0/47, Fa0/48
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

```
R1(config)#do sh vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Gi0/1, Gi0/2, Gi0/3
                                                Gi0/4
10   USERS                            active
20   SERVERS                          active
99   MGMT                             active
999  UNUSED                           active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                                Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24, Fa0/25
                                                Fa0/26, Fa0/27, Fa0/28, Fa0/29
                                                Fa0/30, Fa0/31, Fa0/32, Fa0/33
                                                Fa0/34, Fa0/35, Fa0/36, Fa0/37
                                                Fa0/38, Fa0/39, Fa0/40, Fa0/41
                                                Fa0/42, Fa0/43, Fa0/44, Fa0/45
                                                Fa0/46, Fa0/47, Fa0/48
1002 fddi-default                     act/unsup
1003 token-ring-default               act/unsup
1004 fddinet-default                  act/unsup
 --More--
```

Requirement 4: Redundant core uplinks

Accepted when: Each access switch connects to the core through its own two-link LACP EtherChannel; each trunk carries only lab VLANs, uses VLAN 900 as native and does not negotiate trunking

Status: Met with limitation — SW2 uses LACP; SW3 (3548 XL) has no LACP, so its uplink is a static EtherChannel

> See failure test 1 for EtherChannel summaries and the sanitized configs for the trunk settings.

Evidence and notes

Requirement 5: Access cross-link

Accepted when: A single trunk between SW2 and SW3 provides a backup path to the core

Status: Met

Evidence and notes

```
SW2#show spanning-tree vlan 10

VLAN0010
  Spanning tree enabled protocol ieee
  Root ID    Priority    24586
             Address     0016.9d94.0980
             Cost        12
             Port        56 (Port-channel1)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    28682  (priority 28672 sys-id-ext 10)
             Address     0019.e885.9000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time 300

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Gi0/4               Desg FWD 19        128.4    P2p
Po1                 Root FWD 12        128.56   P2p
```

```
SW3#show spanning-tree vlan 10

Spanning tree 10 is executing the IEEE compatible Spanning Tree protocol
  Bridge Identifier has priority 32768, address 0007.0ecc.6201
  Configured hello time 2, max age 20, forward delay 15
  Current root has priority 24586, address 0016.9d94.0980
  Root port is 2, cost of root path is 12
  Topology change flag not set, detected flag not set, changes 3
  Times:  hold 1, topology change 35, notification 2
          hello 2, max age 20, forward delay 15
  Timers: hello 0, topology change 0, notification 0

Interface Fa0/2 (port 2) in Spanning tree 10 is FORWARDING
   Port path cost 12, Port priority 128
   Designated root has priority 24586, address 0016.9d94.0980
   Designated bridge has priority 24586, address 0016.9d94.0980
   Designated port is 64, path cost 0
   Timers: message age 2, forward delay 0, hold 0
   BPDU: sent 1, received 4004

Interface Fa0/4 (port 16) in Spanning tree 10 is BLOCKING
   Port path cost 19, Port priority 128
   Designated root has priority 24586, address 0016.9d94.0980
   Designated bridge has priority 28682, address 0019.e885.9000
   Designated port is 4, path cost 12
   Timers: message age 11, forward delay 0, hold 0
   BPDU: sent 1, received 875
SW3#
```

We see the cable is being blocked on SW3’s port and is Forwarding on SW2 so the backup port is setup.

Requirement 6: Spanning tree

Accepted when: Rapid PVST+ runs everywhere; R1 is root for every lab VLAN and SW2 is secondary root; the cross-link is blocked on SW3's side; edge ports use PortFast and BPDU Guard

Status: Met with limitation — Rapid PVST+ on R1 and SW2; SW3 (3548 XL) supports only 802.1D

Evidence and notes

```
SW2(config)#do show spanning-tree summary
Switch is in rapid-pvst mode
Root bridge for: VLAN0099
Extended system ID           is enabled
Portfast Default             is disabled
PortFast BPDU Guard Default  is disabled
Portfast BPDU Filter Default is disabled
Loopguard Default            is disabled
EtherChannel misconfig guard is enabled
UplinkFast                   is disabled
BackboneFast                 is disabled
Configured Pathcost method used is short

Name                   Blocking Listening Learning Forwarding STP Active
---------------------- -------- --------- -------- ---------- ----------
VLAN0010                    0         0        0          3          3
VLAN0020                    0         0        0          2          2
VLAN0099                    0         0        0          2          2
VLAN0900                    0         0        0          2          2
---------------------- -------- --------- -------- ---------- ----------
4 vlans                     0         0        0          9          9
```

You can see SW2 in rapid-pvst but now we see that SW2 is the root bridge for vlan 99 when it’s supposed to be R1. To fix we, lower the priority for the vlans on R1 so R1 becomes the root bridge for all the VLANS and the corrected output is here;

```
R1(config)#spanning-tree mode rapid-pvst
R1(config)#end
R1#sh
*Mar  1 11:47:13.921: %SYS-5-CONFIG_I: Configured from console by blake on console
R1#sh spann
R1#sh spanning-tree sum
R1#sh spanning-tree summary | i mode
Switch is in rapid-pvst mode
R1#sh spann
R1#sh spanning-tree sum
R1#sh spanning-tree summary
Switch is in rapid-pvst mode
Root bridge for: VLAN0010, VLAN0020, VLAN0099, VLAN0900
Extended system ID           is enabled
Portfast Default             is disabled
PortFast BPDU Guard Default  is disabled
Portfast BPDU Filter Default is disabled
Loopguard Default            is disabled
EtherChannel misconfig guard is enabled
UplinkFast                   is disabled
BackboneFast                 is disabled
Configured Pathcost method used is short

Name                   Blocking Listening Learning Forwarding STP Active
---------------------- -------- --------- -------- ---------- ----------
VLAN0010                    0         0        0          2          2
VLAN0020                    0         0        0          2          2
VLAN0099                    0         0        0          2          2
VLAN0900                    0         0        0          2          2
---------------------- -------- --------- -------- ---------- ----------
4 vlans                     0         0        0          8          8
R1#
```

Since SW3 is too old for Rapid PVST+ we still will check spanning-tree. With this output we see SW3 see R1 as the root bridge for vlan 99 and that the cross-link is setup correctly because its being blocked on f0/4.

```
SW3#show spanning-tree vlan 99

Spanning tree 99 is executing the IEEE compatible Spanning Tree protocol
  Bridge Identifier has priority 32768, address 0007.0ecc.6203
  Configured hello time 2, max age 20, forward delay 15
  Current root has priority 24675, address 0016.9d94.0980
  Root port is 2, cost of root path is 12
  Topology change flag not set, detected flag not set, changes 6
  Times:  hold 1, topology change 35, notification 2
          hello 2, max age 20, forward delay 15
  Timers: hello 0, topology change 0, notification 0

Interface Fa0/2 (port 2) in Spanning tree 99 is FORWARDING
   Port path cost 12, Port priority 128
   Designated root has priority 24675, address 0016.9d94.0980
   Designated bridge has priority 24675, address 0016.9d94.0980
   Designated port is 64, path cost 0
   Timers: message age 1, forward delay 0, hold 0
   BPDU: sent 76, received 5916

Interface Fa0/4 (port 16) in Spanning tree 99 is BLOCKING
   Port path cost 19, Port priority 128
   Designated root has priority 24675, address 0016.9d94.0980
   Designated bridge has priority 28771, address 0019.e885.9000
   Designated port is 4, path cost 12
   Timers: message age 4, forward delay 0, hold 0
   BPDU: sent 4, received 2853
SW3#
```

Requirement 7: Centralized routing

Accepted when: R1 is the only routing device and the gateway for every VLAN; SW2 and SW3 have only a management address and a default gateway; a VLAN 10 device on SW2 reaches the server on SW3

Status: Met

> Evidence: routing lives only on R1 (see R1-sanitized.txt — SW2/SW3 have only a VLAN 99 address and default gateway). Inter-VLAN reachability is shown in the failure tests (laptop in VLAN 10 reaching 10.10.99.3 through R1).

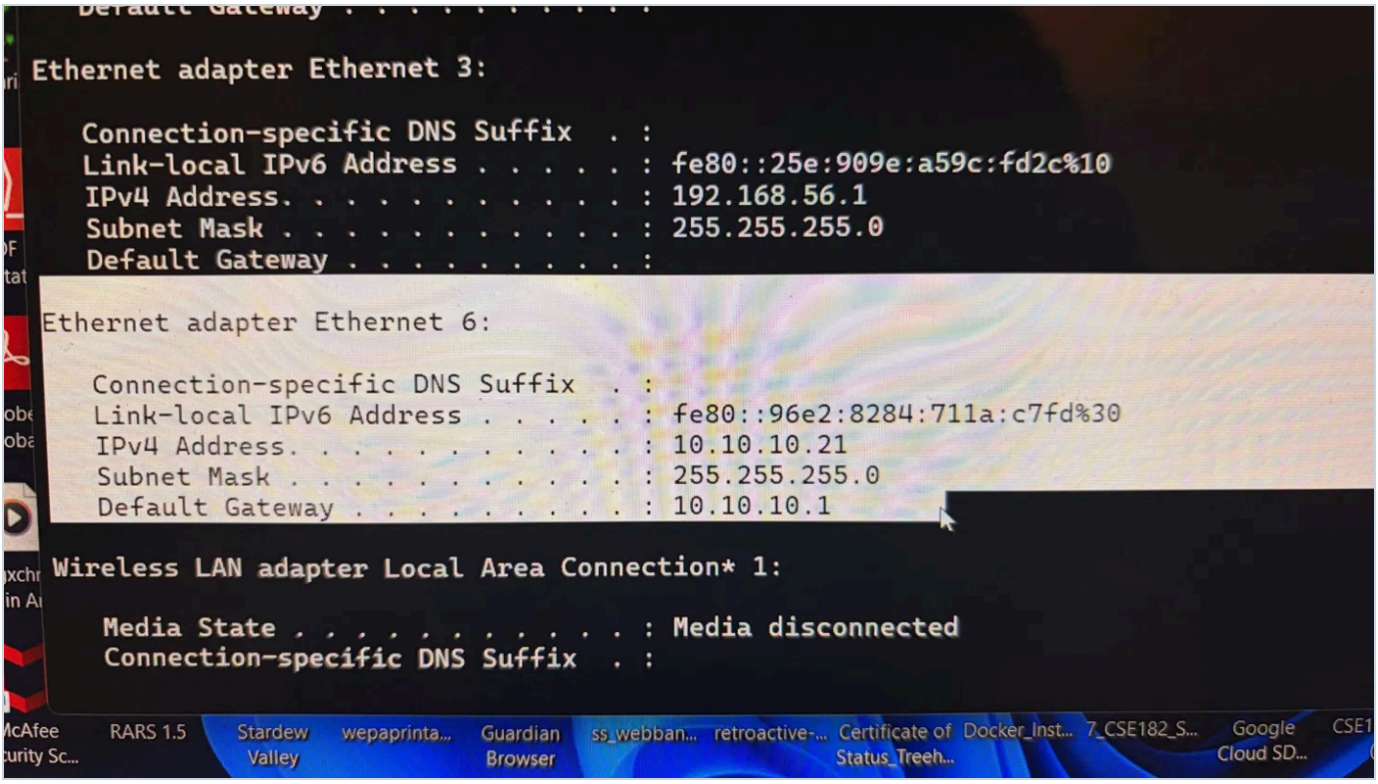
Evidence and notes

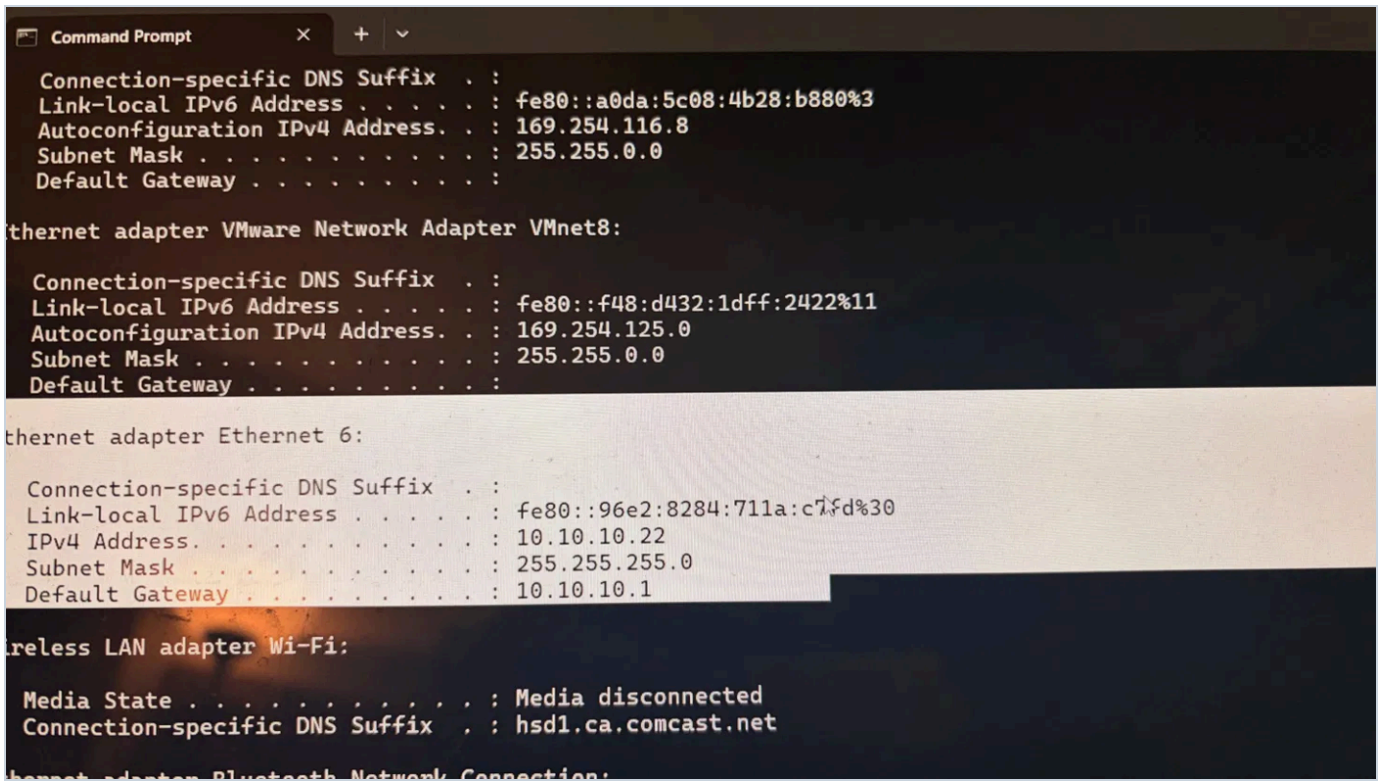
Requirement 8: DHCP

Accepted when: R1 assigns VLAN 10 addresses to devices on either access switch, reserving .1 through .20, with the correct gateway

Status: Met

Evidence and notes



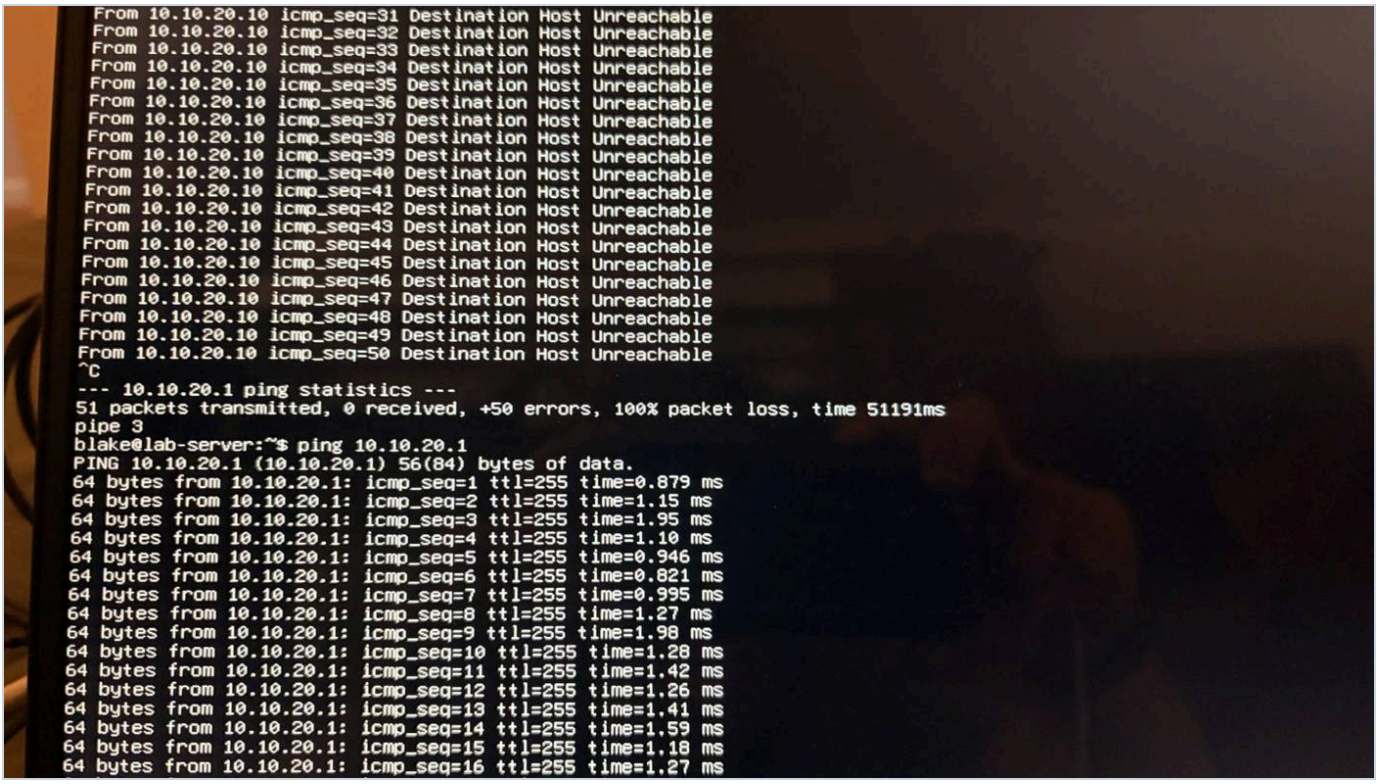

Command Prompt
Connection-specific DNS Suffix . :
Link-local IPv6 Address . . . . . : fe80::a0da:5c08:4b28:b880%3
Autoconfiguration IPv4 Address. . : 169.254.116.8
Subnet Mask . . . . . : 255.255.0.0
Default Gateway . . . . . :
thernet adapter VMware Network Adapter VMnet8:
Connection-specific DNS Suffix . :
Link-local IPv6 Address . . . . . : fe80::f48:d432:1dff:2422%11
Autoconfiguration IPv4 Address. . : 169.254.125.0
Subnet Mask . . . . . : 255.255.0.0
Default Gateway . . . . . :
thernet adapter Ethernet 6:
Connection-specific DNS Suffix . :
Link-local IPv6 Address . . . . . : fe80::96e2:8284:711a:c7fd%30
IPv4 Address. . . . . : 10.10.10.22
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 10.10.10.1
ireless LAN adapter Wi-Fi:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : hsd1.ca.comcast.net

Requirement 9: Logging and backups

Accepted when: The Ubuntu server receives timestamped syslog from all three switches and stores all three running configurations

Status: Met

Evidence and notes

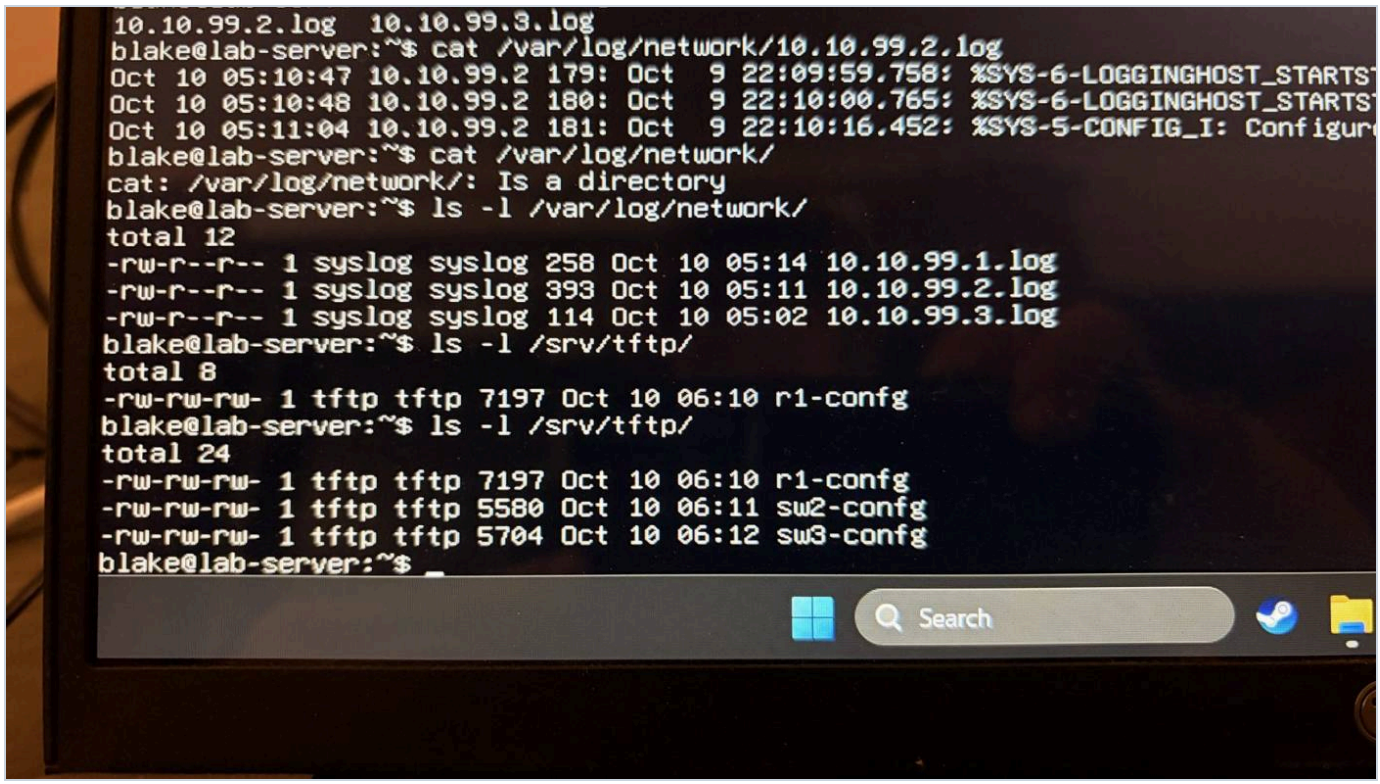


Saved running configs from all 3 devices
and server received them and stored them:

```
R1#copy running-config tftp://10.10.20.10/
Address or name of remote host [10.10.20.10]?
Destination filename [r1-confg]?
!!
7197 bytes copied in 2.541 secs (2832 bytes/sec)
R1#
SW2#copy running-config tftp://10.10.20.10/
Address or name of remote host [10.10.20.10]?
Destination filename [sw2-confg]?
!!
5580 bytes copied in 3.070 secs (1818 bytes/sec)
SW2#SW3, property of Blake Otto

User Access Verification

Username: blake
Password:
SW3>en
Password:
SW3#copy running-config tftp://10.10.20.10/
Address or name of remote host [10.10.20.10]?
Destination filename [sw3-confg]?
!!!
5704 bytes copied in 3.798 secs (1901 bytes/sec)
SW3#
```


10.10.99.2.log 10.10.99.3.log
blake@lab-server:~$ cat /var/log/network/10.10.99.2.log
Oct 10 05:10:47 10.10.99.2 179: Oct 9 22:09:59.758: %SYS-6-LOGGINGHOST_STARTS
Oct 10 05:10:48 10.10.99.2 180: Oct 9 22:10:00.765: %SYS-6-LOGGINGHOST_STARTS
Oct 10 05:11:04 10.10.99.2 181: Oct 9 22:10:16.452: %SYS-5-CONFIG_I: Configur
blake@lab-server:~$ cat /var/log/network/
cat: /var/log/network/: Is a directory
blake@lab-server:~$ ls -l /var/log/network/
total 12
-rw-r--r-- 1 syslog syslog 258 Oct 10 05:14 10.10.99.1.log
-rw-r--r-- 1 syslog syslog 393 Oct 10 05:11 10.10.99.2.log
-rw-r--r-- 1 syslog syslog 114 Oct 10 05:02 10.10.99.3.log
blake@lab-server:~$ ls -l /srv/tftp/
total 8
-rw-rw-rw- 1 tftp tftp 7197 Oct 10 06:10 r1-confg
blake@lab-server:~$ ls -l /srv/tftp/
total 24
-rw-rw-rw- 1 tftp tftp 7197 Oct 10 06:10 r1-confg
-rw-rw-rw- 1 tftp tftp 5580 Oct 10 06:11 sw2-confg
-rw-rw-rw- 1 tftp tftp 5704 Oct 10 06:12 sw3-confg
blake@lab-server:~$
Search

Requirement 10: Internet access

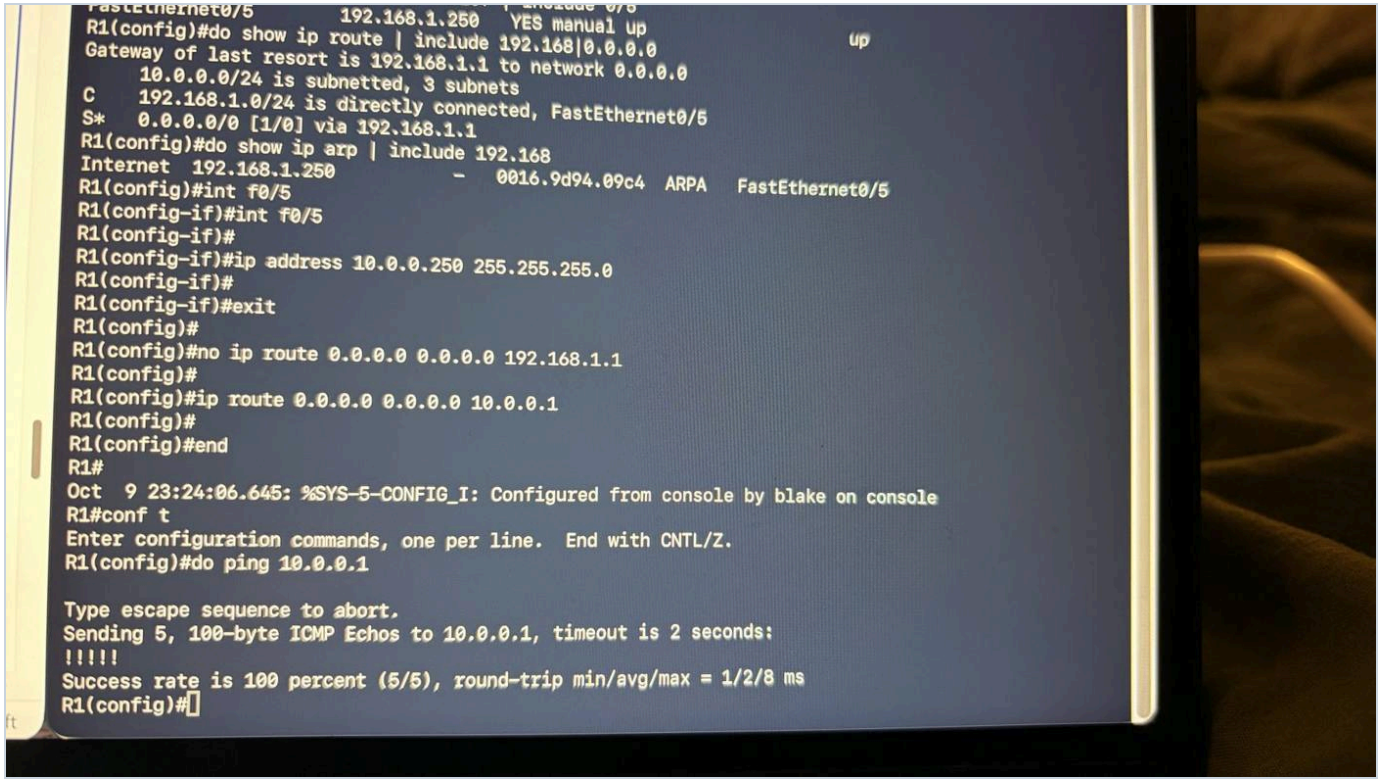
Accepted when: Lab devices reach the internet through the home router over a routed port on R1; lab VLANs and DHCP stay off the home network, and an access list blocks lab devices from home devices

Status: Partially met — documented limitation

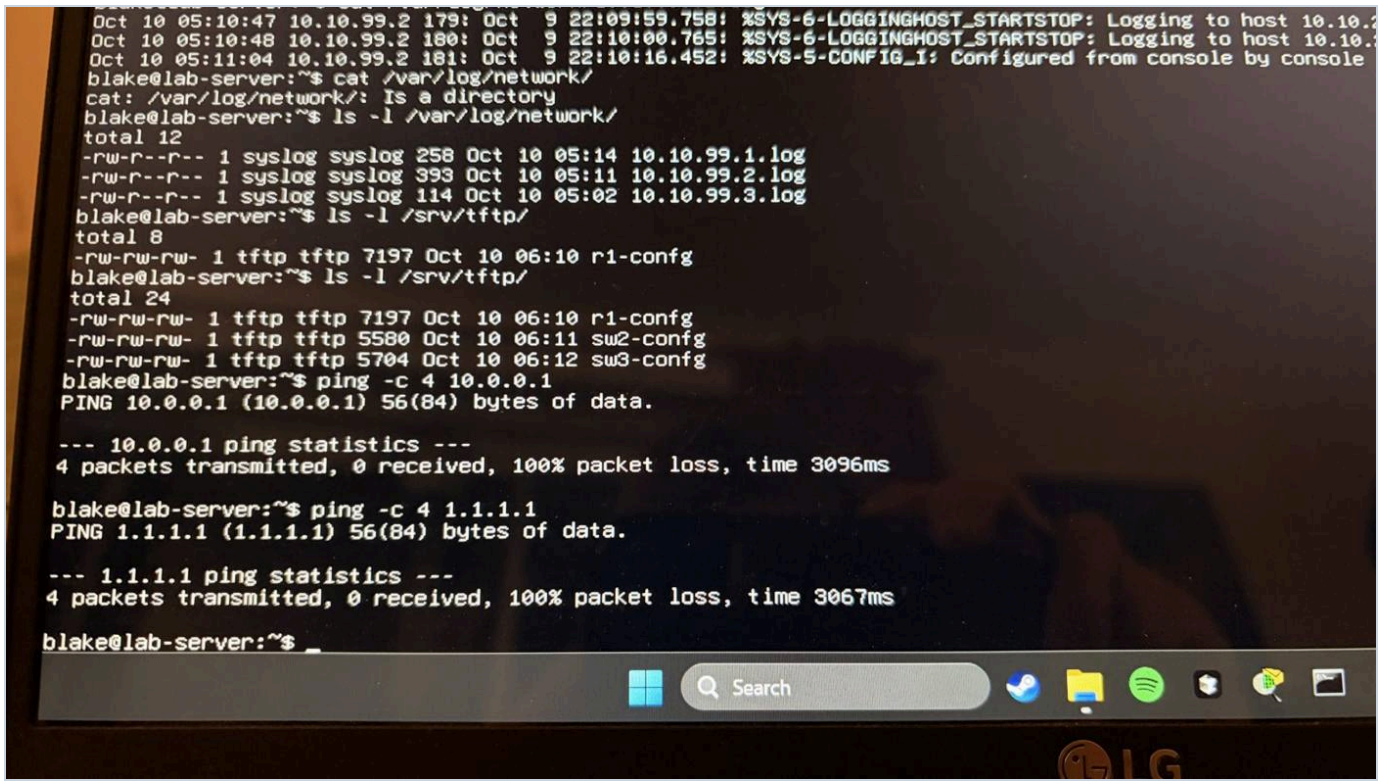
> R1 reaches the internet over its routed uplink, and the ACL blocking lab devices from home devices is in place and tested. Lab hosts cannot reach the internet: the core switch (WS-C3560-48TS) does not support NAT and the ISP gateway does not support static routes, so return traffic has no path. In production, an edge router or firewall performing NAT would solve this.

Evidence and notes

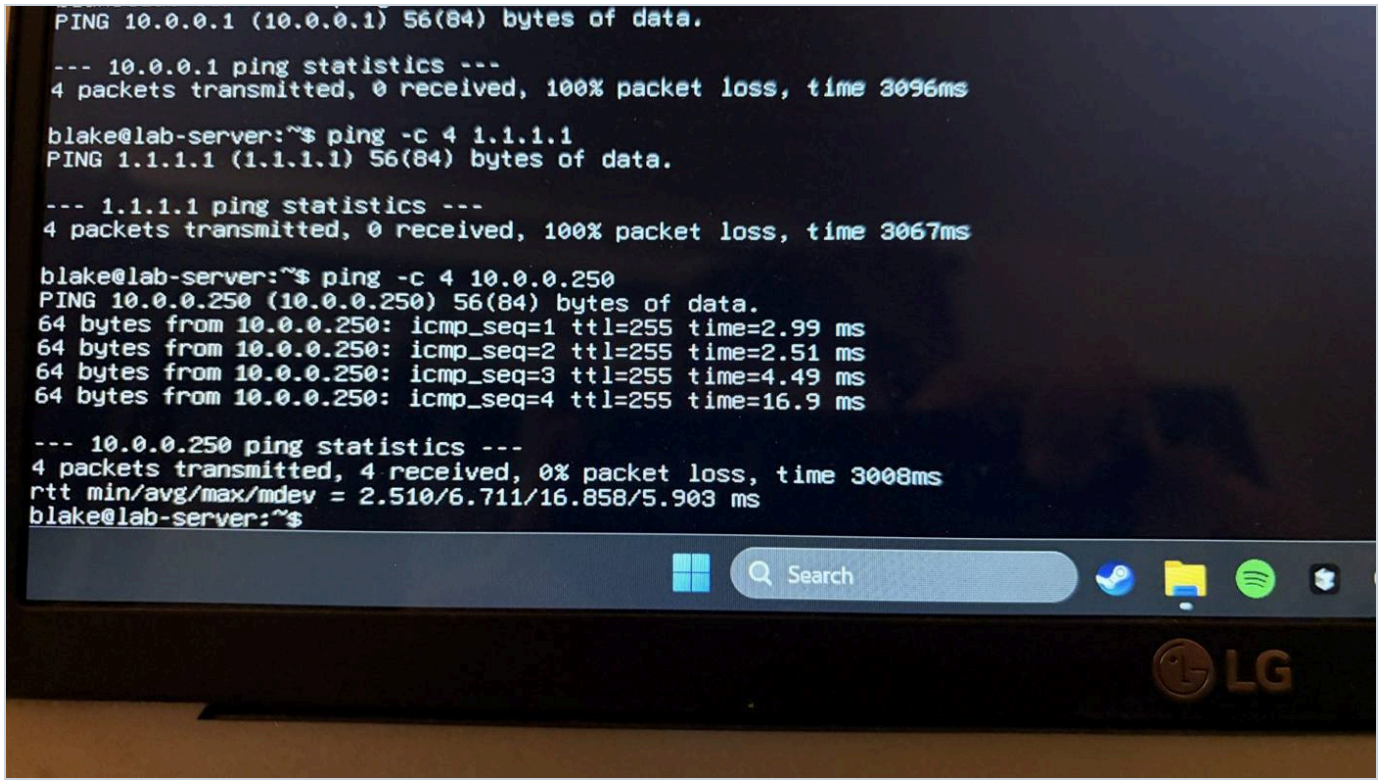
Can’t have lab devices connect to home internet due to static route limitations.



So R1 can ping the home internet.

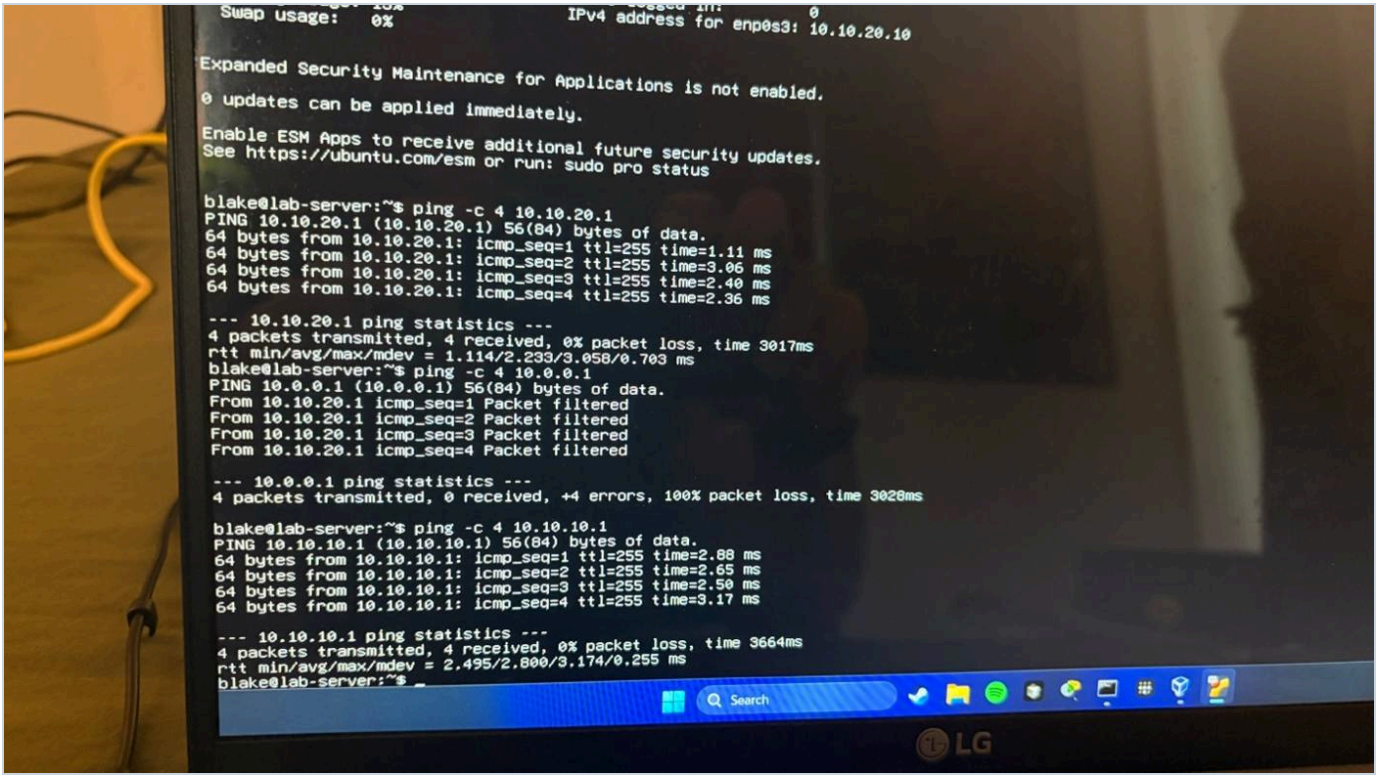


SRV1 can’t get to the internet but can still ping R1.



```
SW3#show mac-address-table int f0/3
Non-static Address Table:
Destination Address  Address Type  VLAN  Destination Port
-------------------  ------------  ----  --------------------
0800.2749.db1d       Dynamic         20  FastEthernet0/3
c8a3.62a2.1bed       Dynamic         20  FastEthernet0/3
SW3#
```

With the image below you can see the ACL filter pings SRV1 and allow certain ip's.




```
R1#   show access-lists LAB_BLOCK_HOME
Extended IP access list LAB_BLOCK_HOME
    10 deny ip 10.10.10.0 0.0.0.255 10.0.0.0 0.0.0.255 log
    20 deny ip 10.10.20.0 0.0.0.255 10.0.0.0 0.0.0.255 log (8 matches)
    30 deny ip 10.10.99.0 0.0.0.255 10.0.0.0 0.0.0.255 log
    40 permit ip any any
R1#   show logging | i IPACCESSLOG
*Mar  1 09:41:37.537: %SEC-6-IPACCESSLOGDP: list LAB_BLOCK_HOME denied icmp 10.10.20.10 -> 10.0.0.1
 (0/0), 1 packet
R1#
```

Requirement 11: Access-layer security (optional)

Accepted when: User ports on both access switches allow one learned device, and DHCP snooping protects VLAN 10 on both, trusting only the paths toward R1

Status: Met on SW2; SW3 (3548 XL) does not support DHCP snooping and has no VLAN 10 user ports

Evidence and notes

To test after I inputted the commands, I tested the ip DHCP snooping on SW2 and we can see the snooping is set up correctly.

```
SW2#show ip dhcp snooping
Switch DHCP snooping is enabled
DHCP snooping is configured on following VLANs:
10
DHCP snooping is operational on following VLANs:
10
DHCP snooping is configured on the following L3 Interfaces:

Insertion of option 82 is disabled
   circuit-id format: vlan-mod-port
    remote-id format: MAC
Option 82 on untrusted port is not allowed
Verification of hwaddr field is enabled
Verification of giaddr field is enabled
DHCP snooping trust/rate is configured on the following Interfaces:

Interface                  Trusted     Rate limit (pps)
------------------------   -------     ----------------
GigabitEthernet0/1         yes         unlimited
GigabitEthernet0/2         yes         unlimited
GigabitEthernet0/4         yes         unlimited
Port-channel1              yes         unlimited
SW2#show port-security interface g0/3
Port Security              : Enabled
Port Status                : Secure-down
Violation Mode             : Shutdown
Aging Time                 : 0 mins
Aging Type                 : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses      : 1
Total MAC Addresses        : 0
Configured MAC Addresses   : 0
Sticky MAC Addresses       : 0
Last Source Address:Vlan   : 0000.0000.0000:0
Security Violation Count   : 0

SW2#
```

Then while plugged in on SW2 g0/3 I renewed my IP address and looked to see if I got an address, which I did, and verified the port security on that interface when connected.

```
SW2#show port-security int g0/3
Port Security              : Enabled
Port Status                : Secure-up
Violation Mode             : Shutdown
Aging Time                 : 0 mins
Aging Type                 : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses      : 1
Total MAC Addresses        : 1
Configured MAC Addresses   : 0
Sticky MAC Addresses       : 1
Last Source Address:Vlan   : c8a3.62a2.1bed:10
Security Violation Count   : 0

SW2#sh ip dhcp snooping binding
MacAddress          IpAddress        Lease(sec)  Type           VLAN  Interface
------------------  ---------------  ----------  -------------  ----  --------------------
C8:A3:62:A2:1B:ED   10.10.10.21      86291       dhcp-snooping   10    GigabitEthernet0/3
Total number of bindings: 1

SW2#show run int g0/3 | i sticky
 switchport port-security mac-address sticky
 switchport port-security mac-address sticky c8a3.62a2.1bed
SW2#
```

I then checked dhcp snooping on SW3 and it cannot do dhcp snooping.

```
SW3(config)#ip dhcp snooping ?
% Unrecognized command
SW3(config)#ip dhcp snooping
```